

(a) Find the speed of P immediately before it collides with Q .

(3)

Immediately after the collision between P and Q , the speed of Q is 5 m s^{-1} .

(b) Show that immediately after the collision P is at rest.

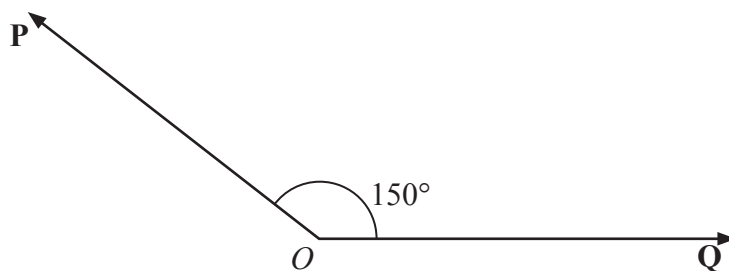
(3)







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- This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.



Two forces **P** and **Q** act on a particle at a point *O*. The force **P** has magnitude 15 N and the force **Q** has magnitude X newtons. The angle between **P** and **Q** is 150° , as shown in Figure 1. The resultant of **P** and **Q** is **R**.

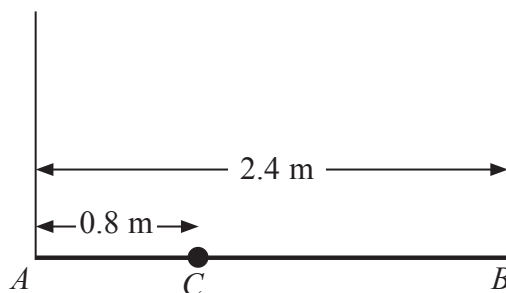
(a) the magnitude of $\mathbf{R}$,

(4)

(b) the value of X .

(5)





A plank AB has mass 12 kg and length 2.4 m . A load of mass 8 kg is attached to the plank at the point C , where $AC = 0.8\text{ m}$. The loaded plank is held in equilibrium, with AB horizontal, by two vertical ropes, one attached at A and the other attached at B , as shown in Figure 2. The plank is modelled as a uniform rod, the load as a particle and the ropes as light inextensible strings.

- (4)

The plank is now modelled as a non-uniform rod. With the new model, the tension in the rope attached at A is 10 N greater than the tension in the rope attached at B .

- (6)

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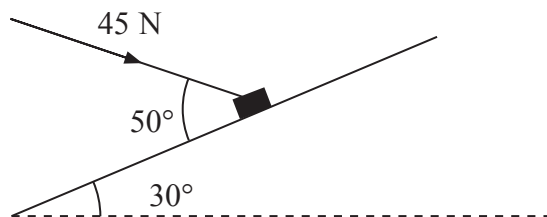


Figure 3

(a) the magnitude of the normal reaction of the plane on the package,

(5)

(b) the coefficient of friction between the plane and the package.

(6)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



(e) Find the time between the instant that the force is removed and the instant that Q comes to rest.

(4)

[illegible]

Question 8 continued



H 2 9 4 9 2 A 0 2 1 2 4